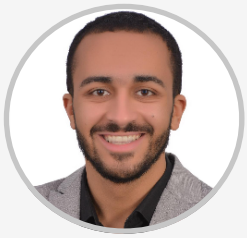
europass



Mostafa Ahmed

Phone number: (+34) 679678934 (Work) | **Email address:** mostafa.ahmed1292002@gmail.com | **Website:** https://github.com/Gaafoor | **LinkedIn:** https://www.linkedin.com/in/mostafa-ahmed-8534141a7/ |

Address: Girona, Spain (Work)

ABOUT ME

Robotics and Intelligent Field Robotic Systems M.Sc. student with experience in SLAM, sensor fusion, computer vision, 3D reconstruction, ADAS, and embedded vehicle systems. Worked with ROS 2, Graph-SLAM, GTSAM, NDT scan matching, EKF, OpenCV, YOLO, COLMAP, Open3D, Python, C/C++, CAN, UDS, AUTOSAR, and STM32/FreeRTOS. Interested in radar-camera fusion, robust localization, semantic mapping, and multi-modal perception for autonomous driving.

EDUCATION AND TRAINING

01/09/2025 – CURRENT
M.SC. IN INTELLIGENT FIELD ROBOTIC SYSTEMS (ERASMUS MUNDUS JOINT MASTER) Joint Degree: Univ. of Girona (UdG), Eötvös Loránd Univ. (ELTE), Univ. of Zagreb (UniZG)

Autonomous Systems, Machine Learning, Multiview Geometry, Planning, Perception, Localization, ADAS, Vehicles & Sensors, and Sensor Fusion.

Website https://ifrosmaster.org/

15/09/2020 – 15/03/2025 Alexandria, Egypt
B.SC. IN MECHATRONICS AND ROBOTICS ENGINEERING Egypt-Japan University of Science and Technology (E-JUST)

Robotics, Control Systems, AI, Embedded Systems, Mechanical Design, and Software Engineering.

Final grade CGPA 3.35 / 4.00 (Very Good)

WORK EXPERIENCE

SPACE ROBOTICS AND GNC INTERN – ADVANCED CENTER FOR AEROSPACE TECHNOLOGIES (CATEC) – 01/06/2026 – 31/08/2026 – SEVILLA, SPAIN

- Developed and evaluated spacecraft robotics simulation workflows for Guidance, Navigation, and Control (GNC), docking, rendezvous, contact dynamics, optical navigation, and autonomy.
- Built Software-in-the-Loop (SIL) benchmark scenarios using Gazebo, MuJoCo, NVIDIA Isaac Sim, Basilisk, and Space ROS for robotic manipulation and spacecraft control validation.
- Executed mission design and cross-tool physics benchmarking across GMAT, FreeFlyer, NASA 42, Basilisk, and Orekit, producing metrics, plots, and technical reports for simulation tool selection.

RESEARCH ASSISTANT – CONTROL LAB, EGYPT-JAPAN UNIVERSITY OF SCIENCE AND TECHNOLOGY (E-JUST) – 01/09/2024 – 28/02/2025 – ALEXANDRIA, EGYPT

- **FPGA Integration:** Developed a LabVIEW I2C subroutine for NI sbRIO FPGA to evaluate mechatronic sensor performance.
- **Academic Research:** Co-authored a comparative study on sensor costs, currently under peer review.

TEAM LEADER – E-JUST RACING TEAM (STUDENT ENGINEERING TEAM) – 01/04/2022 – 01/02/2025 – EGYPT

- **Leadership & EV Systems**: Led cross-functional teams in EVER and Shell Eco-marathon, mentored junior engineers, and directed electric vehicle design/manufacturing, achieving 6th place in EVER.

ELECTRICAL AUTOMATION ENGINEER INTERN – NESTLÉ – 15/07/2023 – 15/09/2023 – GIZA, EGYPT

- **Automation**: Programmed Allen-Bradley PLCs, integrated OPC tags with DMO platform, and supported hardware retrofits for 40+ obsolete components, reducing downtime by 20%.

SKILLS

Robotics & Autonomy

ROS 2 | RViz | SLAM | EKF | sensor fusion | Teleoperation | mobile manipulation | motion planning | Trajectory planning | GTSAM | NDT Scan Matching

Simulation & Testing

MATLAB/Simulink | Gazebo | MuJoCo | NVIDIA Isaac Sim | LabVIEW | SolidWorks | Software-in-the-Loop (SIL)

Computer Vision & 3D Mapping

OpenCV | YOLO | ArUco | COLMAP | Structure-from-Motion | Multi-View Stereo | Open3D | 3D Reconstruction | Semantic Segmentation

Programming

Python | C/C++ | Linux | NumPy | Git

Vehicle Systems & Communication

ADAS | CAN | I2C | SPI | UDS ISO14229 | AUTOSAR | LKA | ESP8266 IoT | FOTA

Embedded Systems

STM32 | Raspberry Pi | FreeRTOS | Sensor Interfacing

PROJECTS

01/02/2026 – 30/06/2026
Graph-SLAM Localization & Kinodynamic Exploration for TurtleBot 4

- Implemented a ROS 2 Jazzy autonomy stack combining covariance-triggered Graph SLAM, GTSAM iSAM2 pose-graph optimization, NDT scan matching, loop-closure detection, and RViz-based uncertainty visualization.
- Integrated optimized Graph-SLAM localization with frontier exploration, kinodynamic RRT planning, occupancy mapping, and pure-pursuit path following for TurtleBot 4 navigation.
- **Tools**: ROS 2 Jazzy, C++, Python, GTSAM, PCL, Eigen, RViz, slam_toolbox, TurtleBot 4.

01/02/2026 – 30/06/2026
Underwater Structure-from-Motion & Factor-Graph Reconstruction

- Developed an underwater perception pipeline for Girona 500 / Stonefish AUV data, combining COLMAP visual reconstruction, loop-closure-aware keyframe selection, and GTSAM factor-graph optimization over 3,641 odometry frames and 279 visual keyframes.
- Fused noisy DVL/INS odometry with robust visual reprojection constraints, reducing mean reprojection error from 22.1 px to 0.47 px and generating a dense Poisson mesh with 3.67M fused points and 10.6M vertices.
- **Tools**: Python, ROS bags, COLMAP, GTSAM, Open3D, NumPy, Structure-from-Motion, Multi-View Stereo.

11/2023 – 02/2024
Enhancing AUTOSAR FOTA Updates with Practical Steering Implementation

- **Connected Vehicle Systems**: Developed an AUTOSAR-based STM32/FreeRTOS FOTA system with CAN, UDS authentication, SPI-WiFi communication, and memory stack implementation.
- **ADAS Prototype**: Integrated an OpenCV-based lane-keeping component with embedded steering control to demonstrate practical vehicle automation.
- **System Impact**: Implemented delta updating with **51.6%** size reduction and real-time multitasking for connected automotive software updates.
- **Tools**: STM32CubeIDE, FreeRTOS, CAN, UDS, SPI-WiFi, ESP8266, OpenCV, AUTOSAR.

Links https://github.com/Gaafoor/FOTA-and-Remote-Diagnostics | https://drive.google.com/file/d/1Kdqgw9tU31s-YV1I_Me6czNDnHsGwFPG/view?usp=sharing

17/12/2025
Microscopy Hackathon – Deep Learning for Atomic Defect Analysis

- **Semantic Segmentation:** Co-developed a U-Net-based deep learning pipeline for 6-class atomic segmentation in HAADF-STEM microscopy images of 2D transition metal dichalcogenides, achieving 0.94 F1-score for atom detection.
- **Physics-Informed Classification**: Integrated AtomAI-based atomic patch classification with crystallographic constraints, improving chemical species classification accuracy from ~65% to 94.2%.

- **Unsupervised Anomaly Detection**: Applied self-supervised autoencoder features, atom tracking, and spatial density analysis to identify dynamically unstable atomic sites and defect-rich lattice regions.
- **Tools**: Python, U-Net, AtomAI, semantic segmentation, self-supervised learning, autoencoders, anomaly detection, HAADF-STEM microscopy.

Link https://kaliningroup.github.io/mic_hackathon_2/submissions/

03/2023 – 10/2023
Minesweepers – Autonomous Vehicle-Manipulator System

- **Global Award**: Secured 2nd place globally by developing a ROS-based vehicle-manipulator robot for surface landmine detection, localization, and collection.
- **Manipulation**: Integrated a robotic arm with an electromagnetic end-effector for safe mine pickup and multi-point collection.
- **Perception & Control**: Implemented YOLO detection, EKF localization, IMU fusion, teleoperation, RViz visualization, and a custom mapping GUI.
- **Tools**: ROS, Python, Raspberry Pi, OpenCV, YOLO, RViz, EKF, IMU.

Link https://drive.google.com/file/d/1FyOOmM8Gu7zhUWVCsp3qW0ShII_X-BxA/view?usp=drive_link

01/02/2026 – 30/06/2026
Vehicle-Manipulator System

- Built a ROS 2 mobile manipulation system for a TurtleBot-based robot with a SwiftPro arm, integrating kinematic modeling, forward kinematics validation, resolved-rate motion control, and weighted damped least-squares inverse kinematics.
- Implemented recursive task-priority control for constrained manipulation, including end-effector tracking, joint-limit handling, base-arm coordination, and configurable task hierarchies.
- Developed an FSM-based pick-and-place/intervention pipeline using ArUco marker perception, TF2 frame transforms, gripper control, and ROS 2 visualization/debugging tools.
- **Tools**: ROS 2, Python, NumPy, TF2, ArUco, inverse kinematics, task-priority control, mobile manipulation.

HONOURS AND AWARDS

03/2023
1st Place Global Winner, Lunar Lander Extended Reality Challenge (ASME)

Link https://drive.google.com/file/d/101gxYfzW9zlZOfWYiZump0Pv5-DdXAnq/view?usp=sharing

PUBLICATIONS

2025
Enhancing AUTOSAR-Based Firmware Over-the-Air Updates in the Automotive Industry with a Practical Implementation on a Steering System

Formal publication detailing the **Valeo Best Project Award-winning** secure FOTA system.

Bachelor's Graduation Thesis

Authors: Mostafa A. Mostafa, et al. | **Publisher**: Egypt-Japan University of Science and Technology

2024
Automating Hot-Rolling: Designing an Integrated Mechatronics System for Enhanced Efficiency in Sheet Metal Production

Modeled multi-body dynamics (AC motor, power screw) in **MATLAB/Simulink** and designed a **PID controller** to automate sheet metal hot-rolling speed and thickness.

Authors: Mostafa A. Mostafa, et al.

2023
Implementing PID Controller on a Hand Follower Robot Using Numerical Methods

Developed a **PID controller** using **numerical methods** and **auto-tuning** techniques for a hand-follower robot to optimize system stability and eliminate steady-state error.

Authors: Mostafa A. Mostafa, et al.